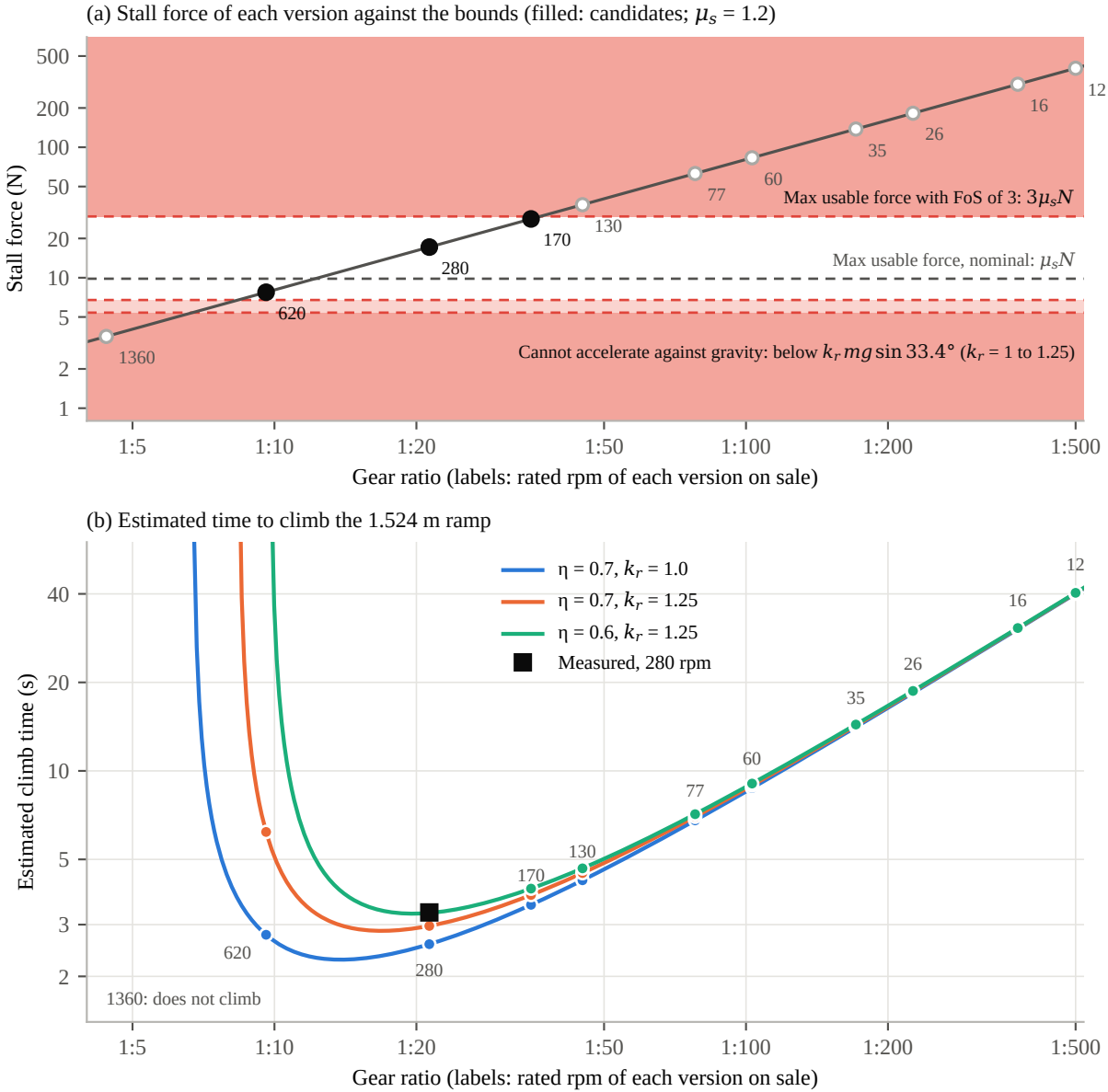
(a) Stall force of each version against the bounds (filled: candidates; μ_s = 1.2)
Stall force (N)
500
200
100
50
20
10
5
2
1
1360
620
280
170
130
77
60
35
26
16
12
Max usable force with FoS of 3: $3\mu_s N$
Max usable force, nominal: $\mu_s N$
Cannot accelerate against gravity: below $k_r\, mg \sin 33.4°$ (k_r = 1 to 1.25)
1:5
1:10
1:20
1:50
1:100
1:200
1:500
Gear ratio (labels: rated rpm of each version on sale)
(b) Estimated time to climb the 1.524 m ramp
Estimated climb time (s)
40
20
10
5
3
2
η = 0.7, k_r = 1.0
η = 0.7, k_r = 1.25
η = 0.6, k_r = 1.25
Measured, 280 rpm
620
280
170
130
77
60
35
26
16
12
1360: does not climb
1:5
1:10
1:20
1:50
1:100
1:200
1:500
Gear ratio (labels: rated rpm of each version on sale)